CS486 – Introduction to Databases

LAB

Topic 0: Setup Guide

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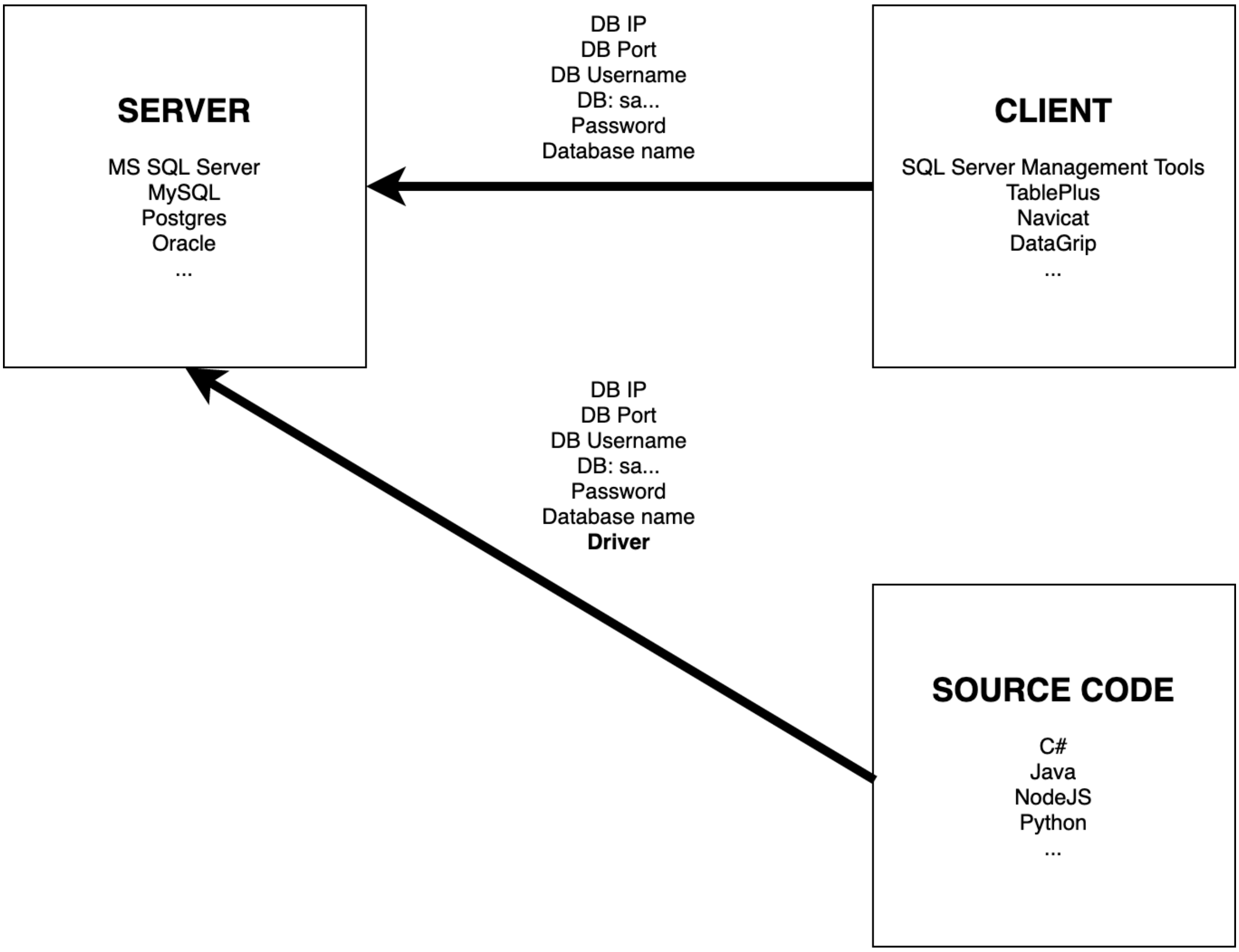
Setup Guide

1. Notes

Please install the tools before coming to the lab hour.

In case you have troubles, please note them and ask during the lab hour.

2. Downloads



SQL Server: https://www.microsoft.com/en-us/sql-server/sql-server-downloads

SQL Management Studio:
https://learn.microsoft.com/en-us/ssms/download-sql-server-management-studio-ssms

Docker CE (Mac/Linux): https://docs.docker.com/engine/install/

Visual Studio Code (Mac/Linux): https://code.visualstudio.com/docs/setup/setup-overview

TablePlus (Mac optional): https://docs.tableplus.com/

3. Install SQL Server instance

There are 2 ways to install SQL Server instance:

- SQL Server for Windows: Only applicable for Windows users
- SQL Server for Linux: Applicable for all platforms such as Linux, Mac, Windows via Docker

You can freely choose any way to use SQL Server in most lab assignments with an exception that some assignments can only be done on Windows (SQL Management Studio), please complete them in the lab session. It is recommended that you should try both ways.

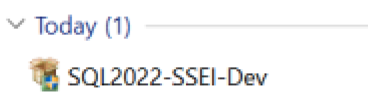
3.1. SQL Server for Windows (2022)

In this tutorial, you will learn to install the SQL Server 2022 Developer Edition and SQL Server Management Studio (SSMS)

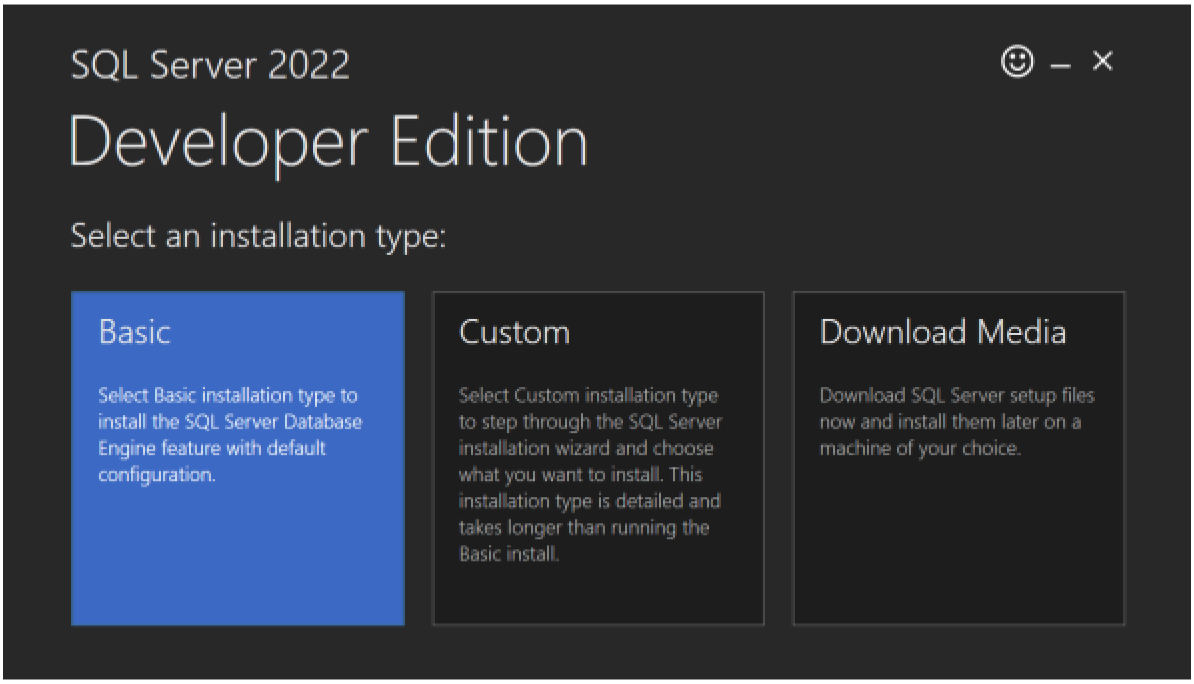
1. Install SQL Server 2022 Developer Edition

- Download sql server 2022 Developer Edition

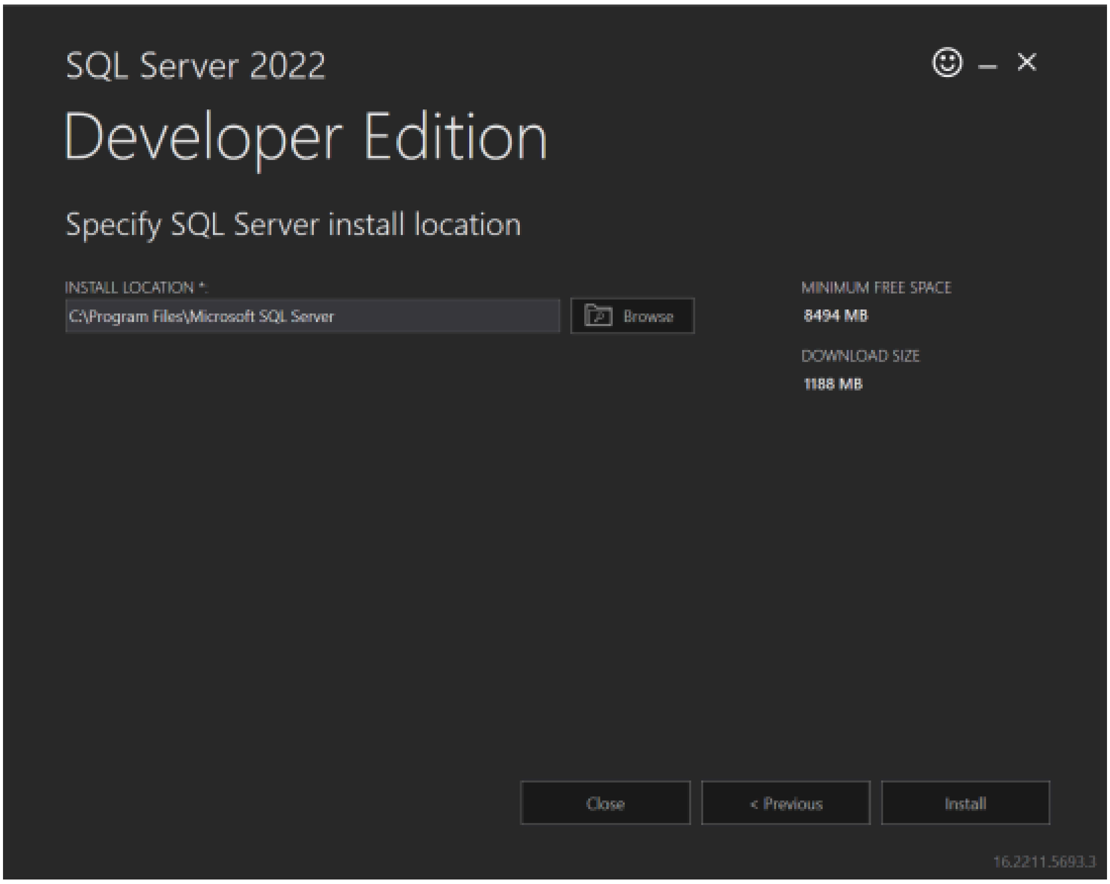
- Double-click the file SQL2022-SSEI-Dev.exe file to launch the downloader.



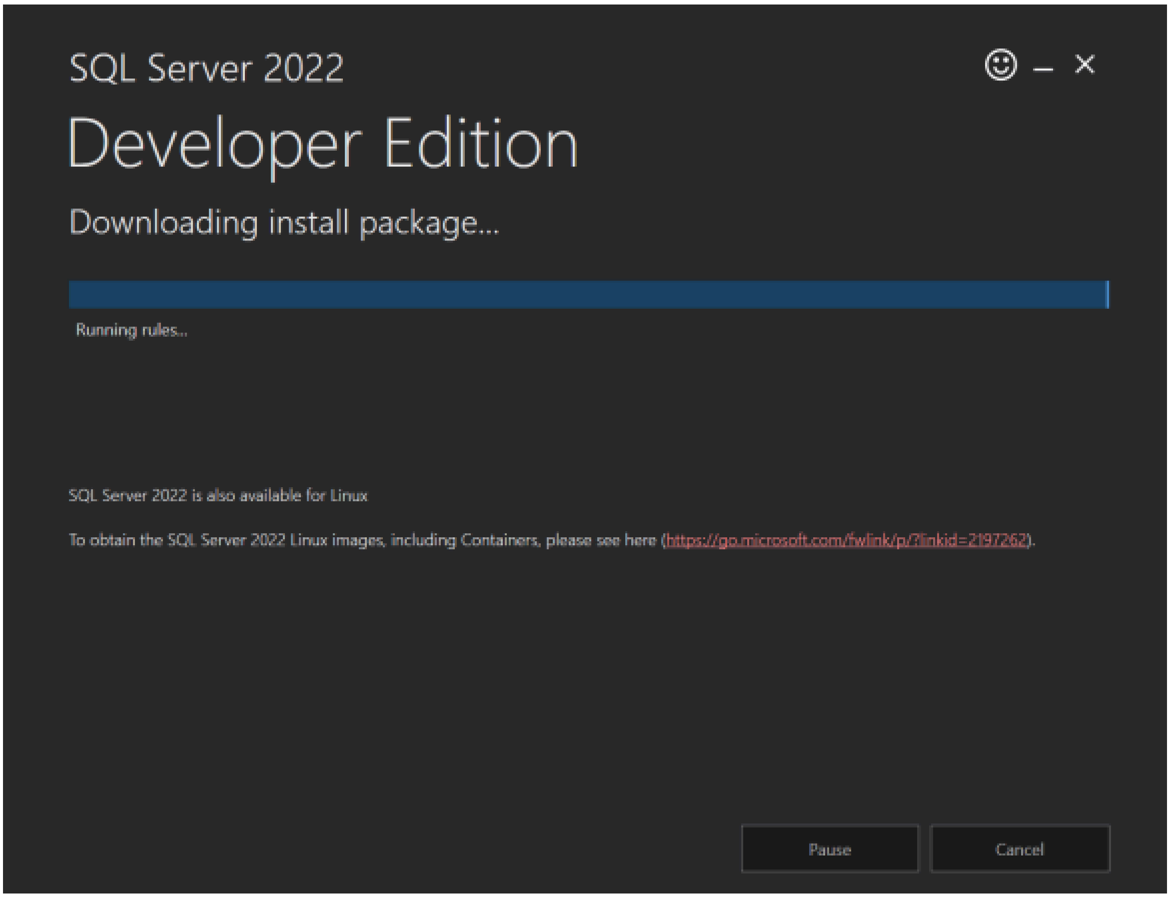
- The downloader will ask you to select the installation type, choose Basic.



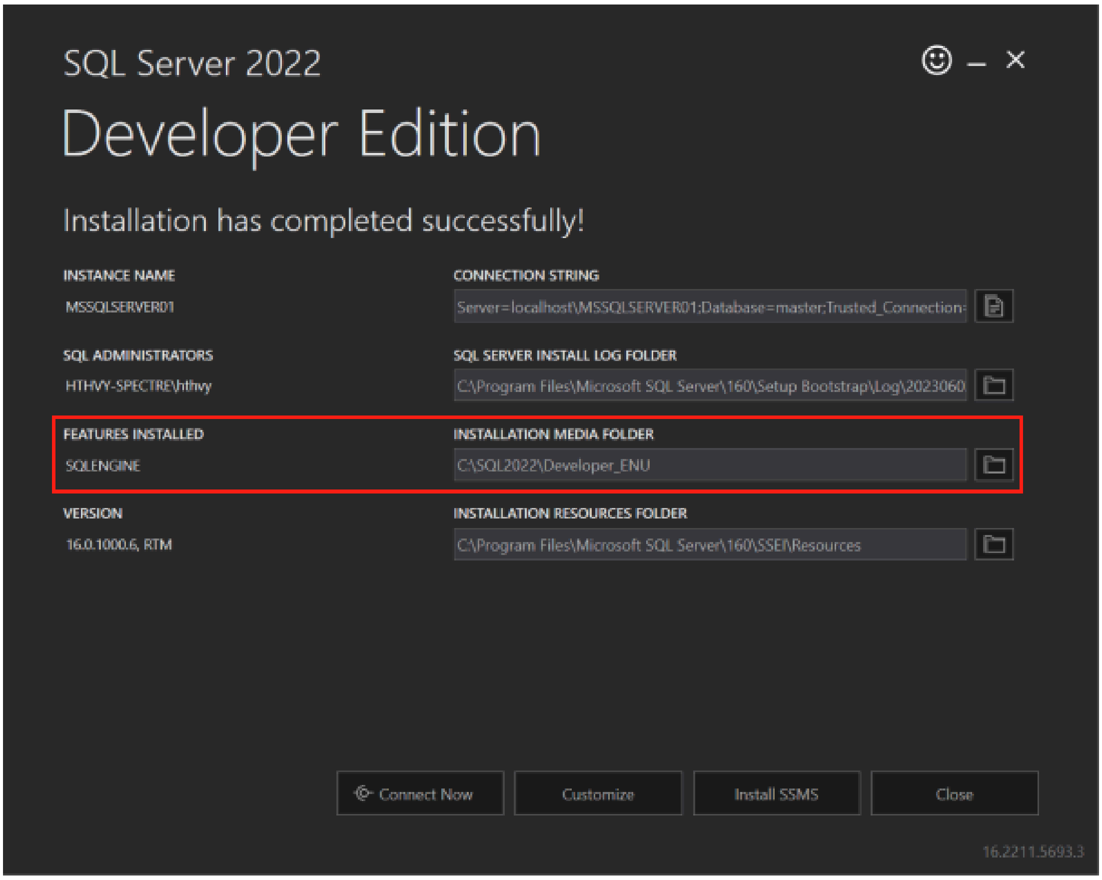
- Click Accept Button > Specify SQL server install location.



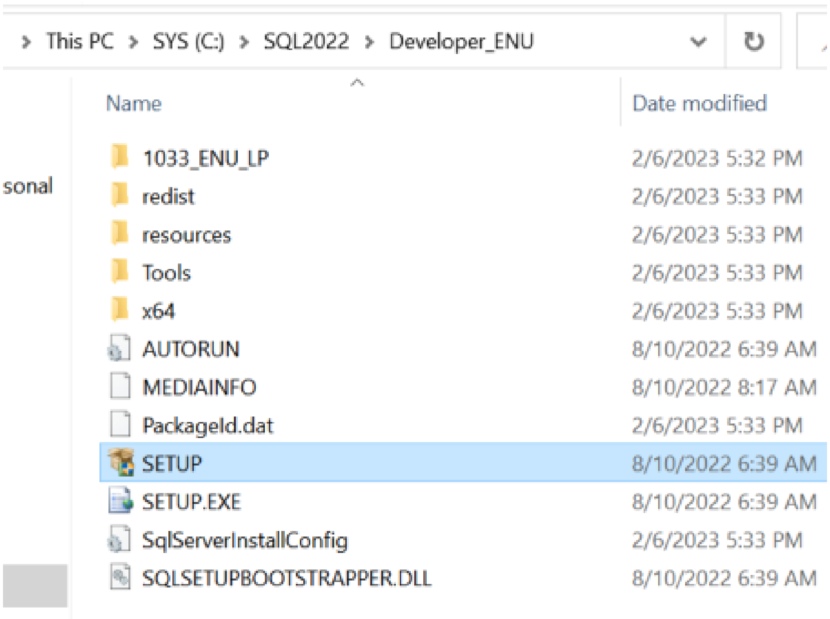
- Then click the Install button.
- The downloader will start downloading the installation files. It'll take a while.



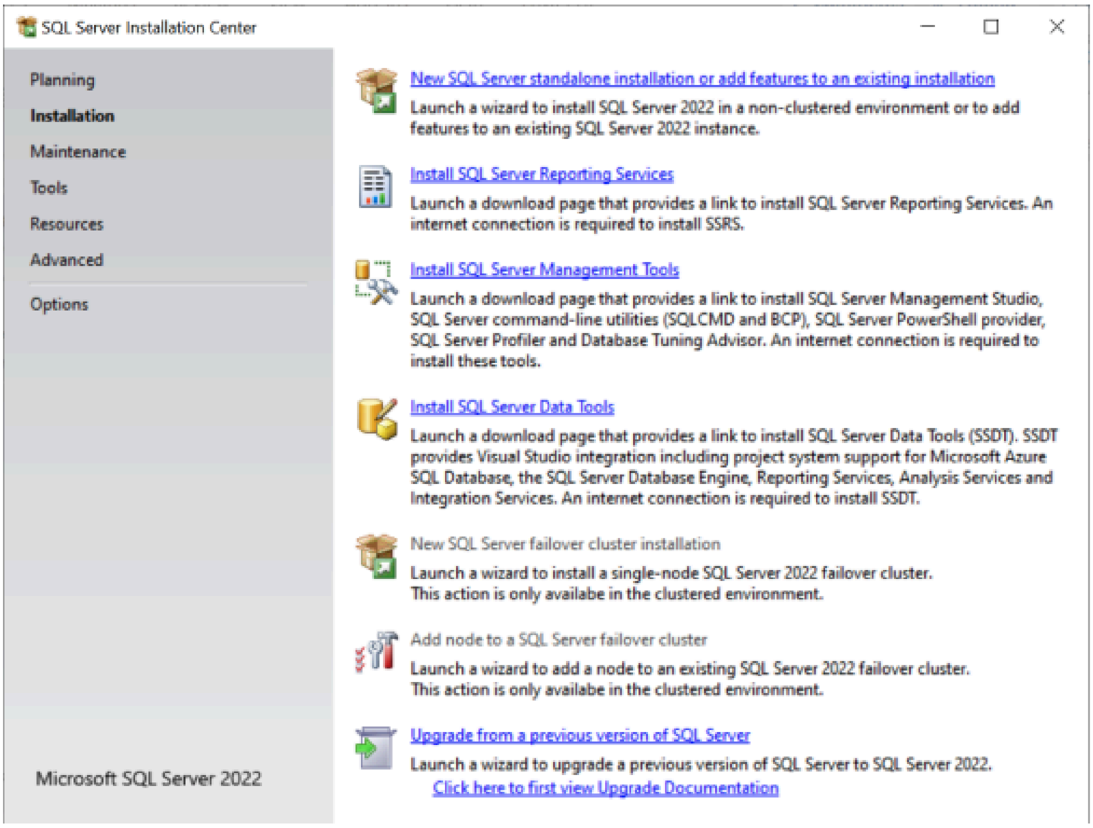
- Once the download completes, open the folder that stores the downloaded file.



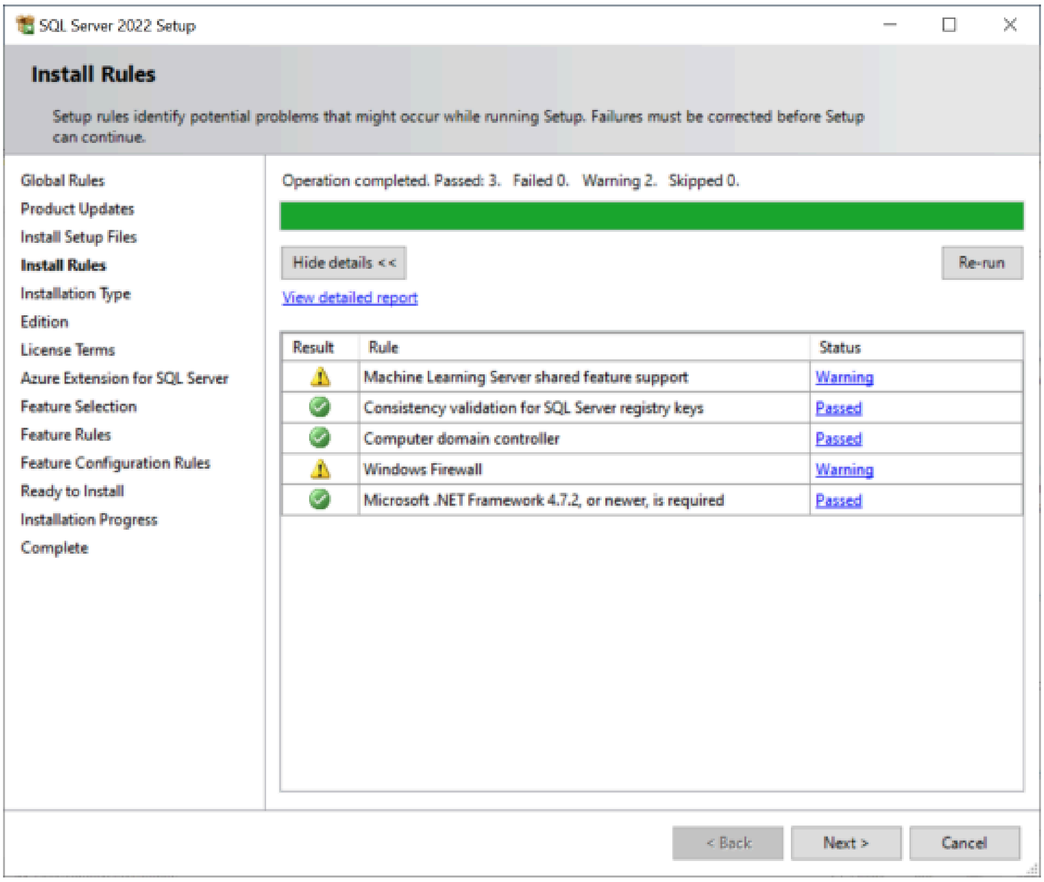
- Open the installation folder and click the setup.exe file to launch the installer.



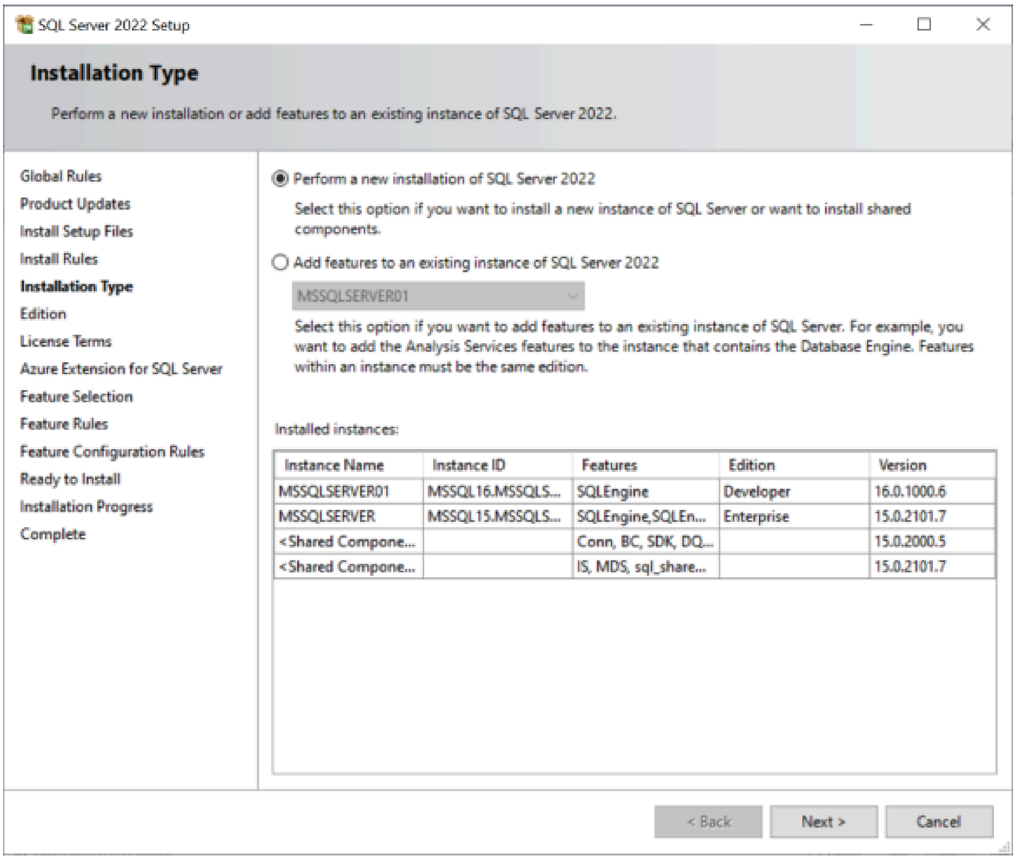
- Install SQL Server 2022 developer edition
  - After double click setup.exe, you'll see the following window; select the installation option on the left:



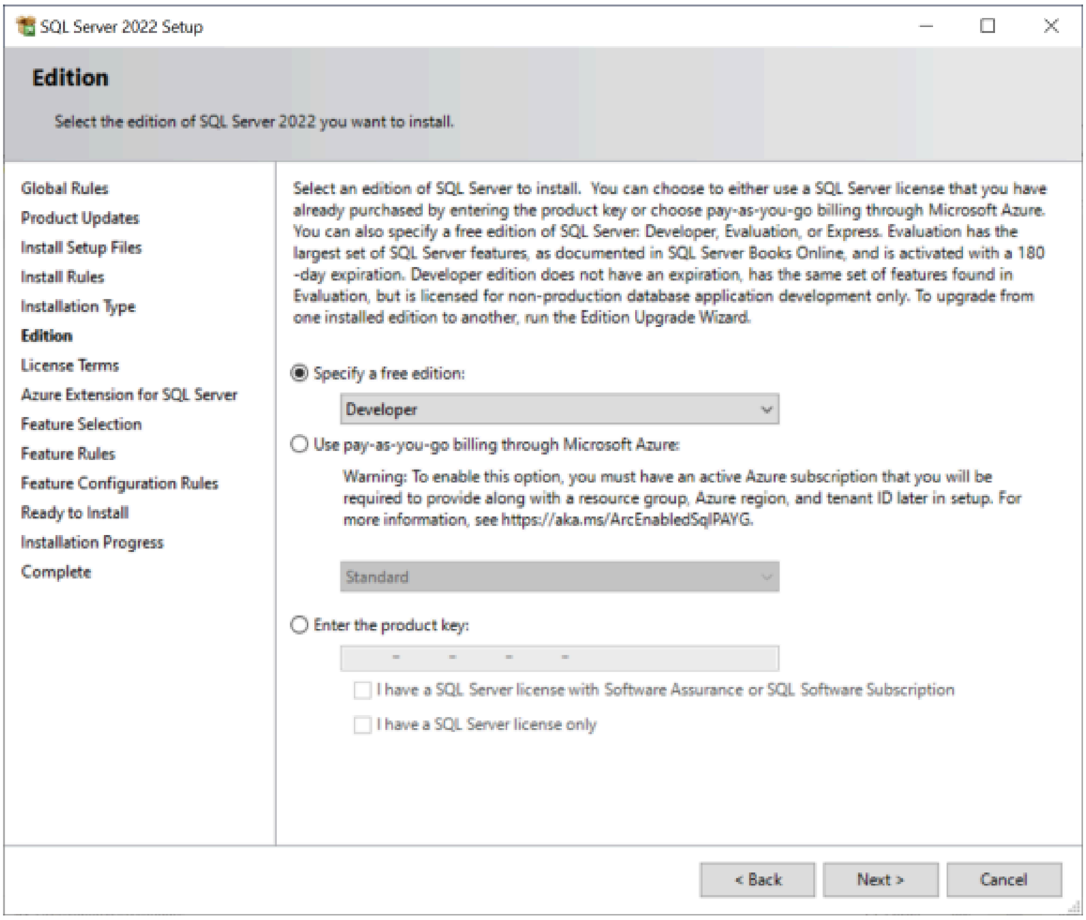
- Click the first link to launch a wizard to install SQL Server 2022 > Next.



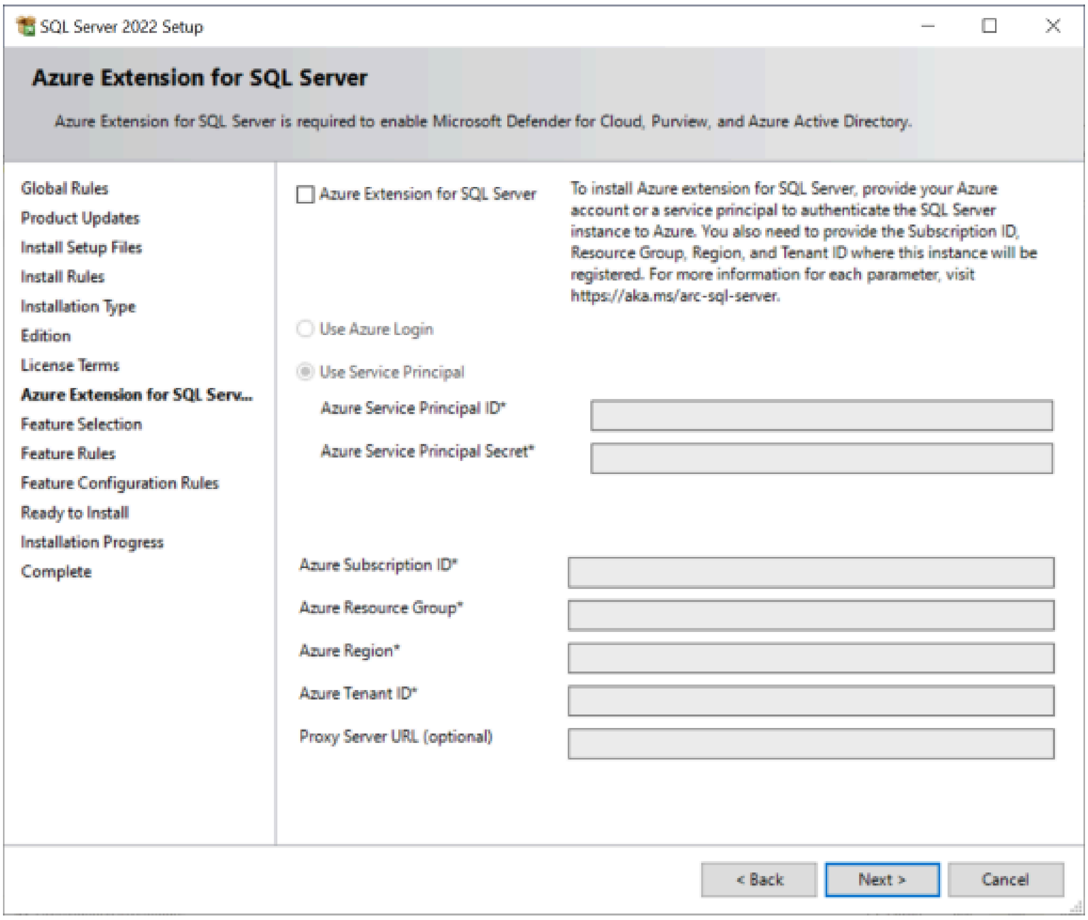
- Click Next > New installation



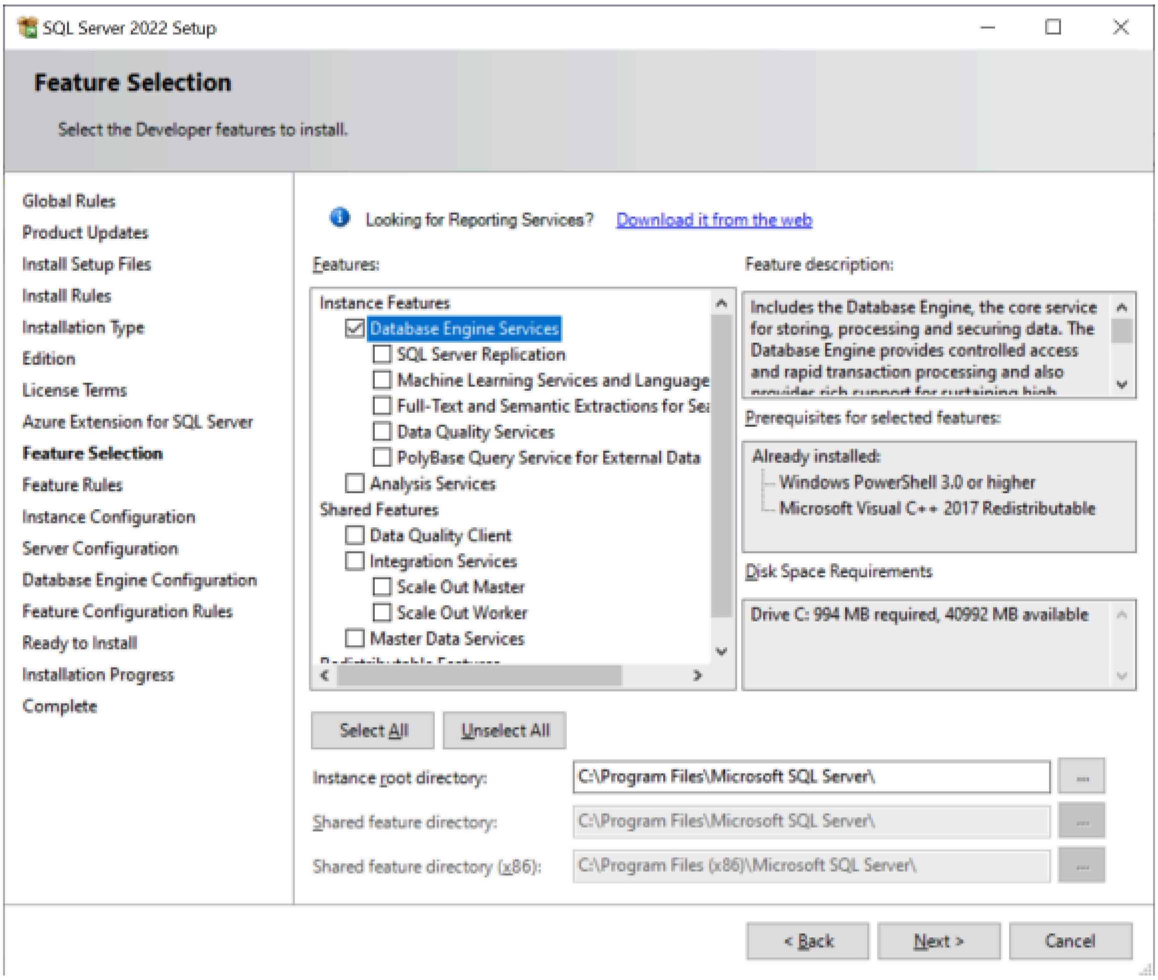
- Specify the edition that you want to install, select Developer edition, and click the Next button



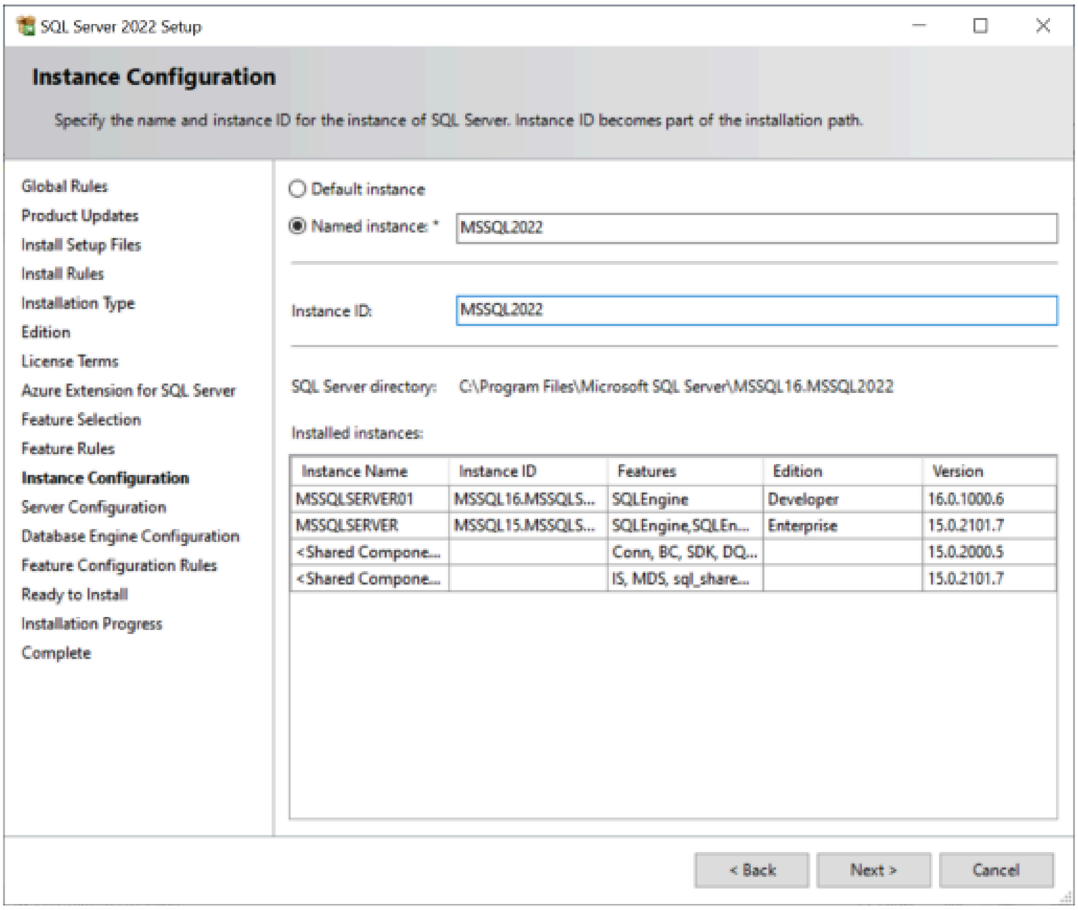
- Select the "I accept the license terms." then click the Next button.
- Uncheck the "Azure extension for SQL server" if you don't want and click the Next button



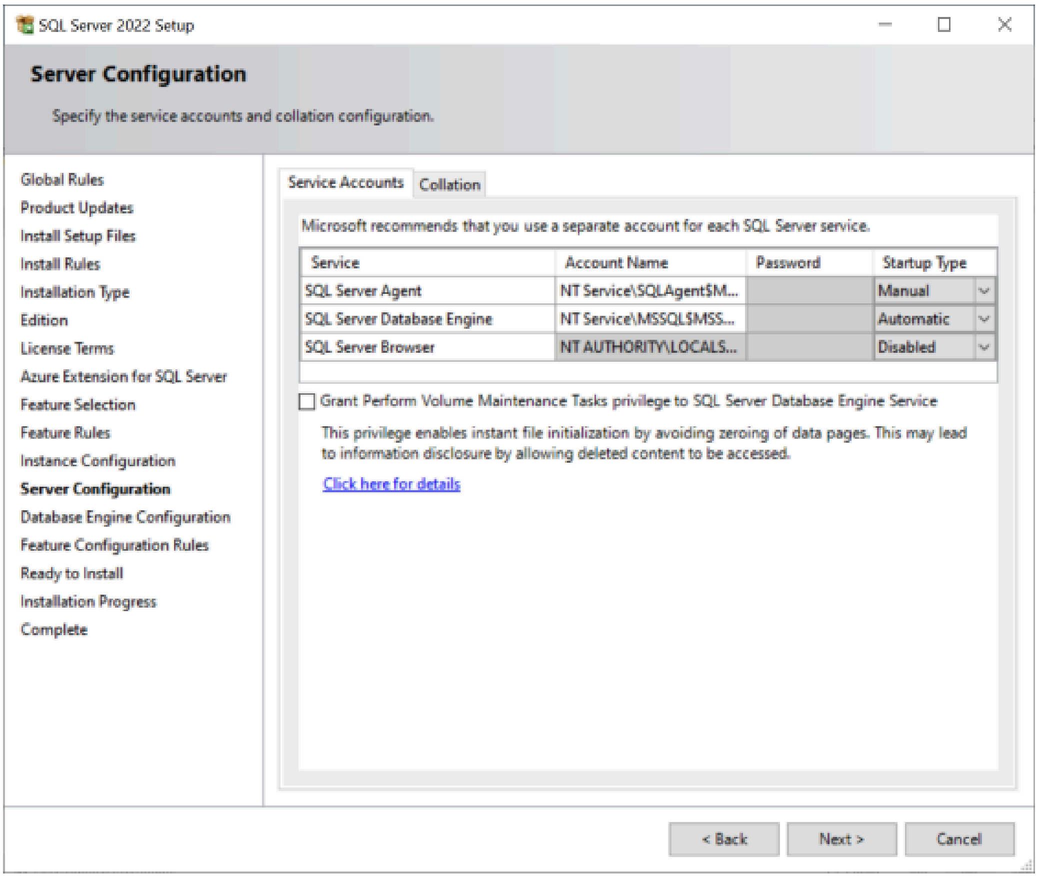
- Select the features that you want to install. For learning purposes, you need the Database Engine Services; check the checkbox and click the Next button to continue.



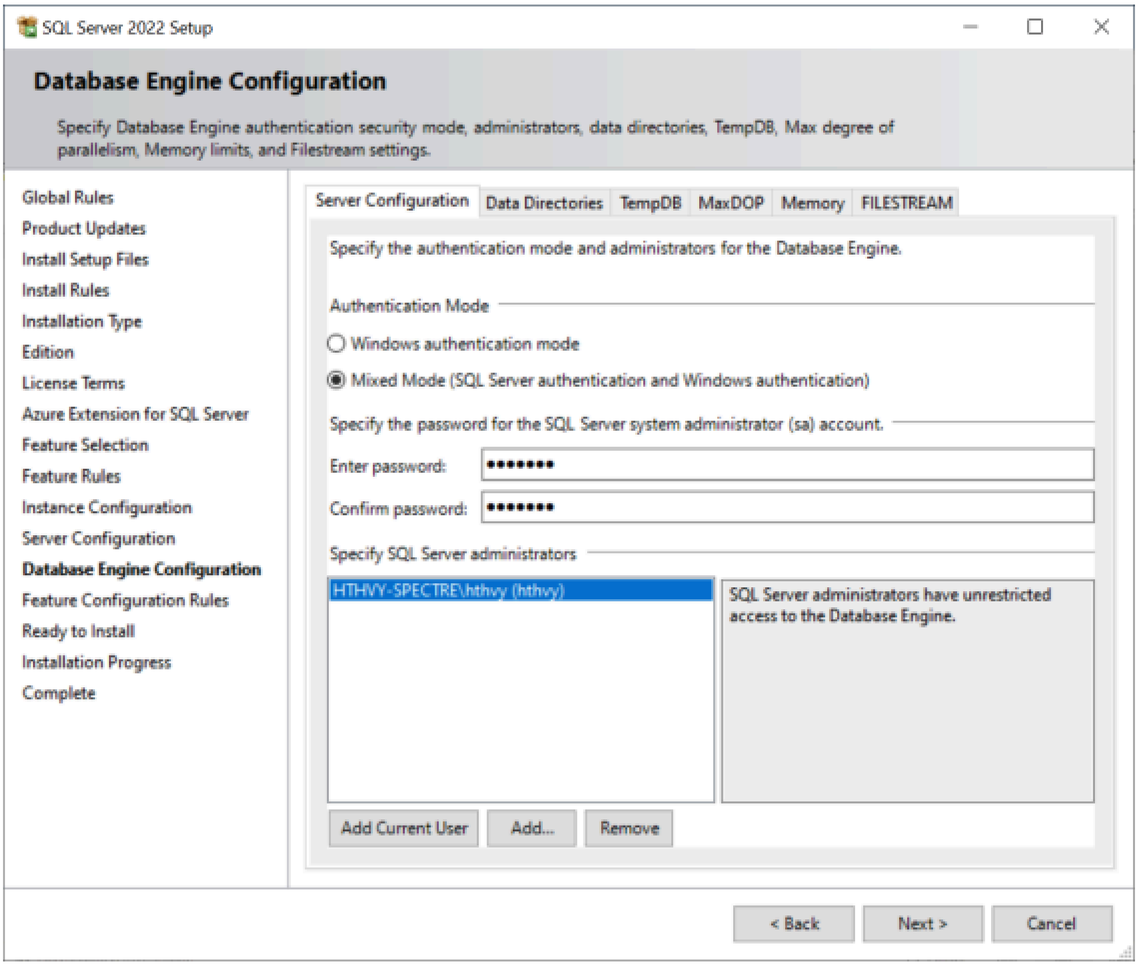
- Provide the instance ID of the SQL Server and click the Next button



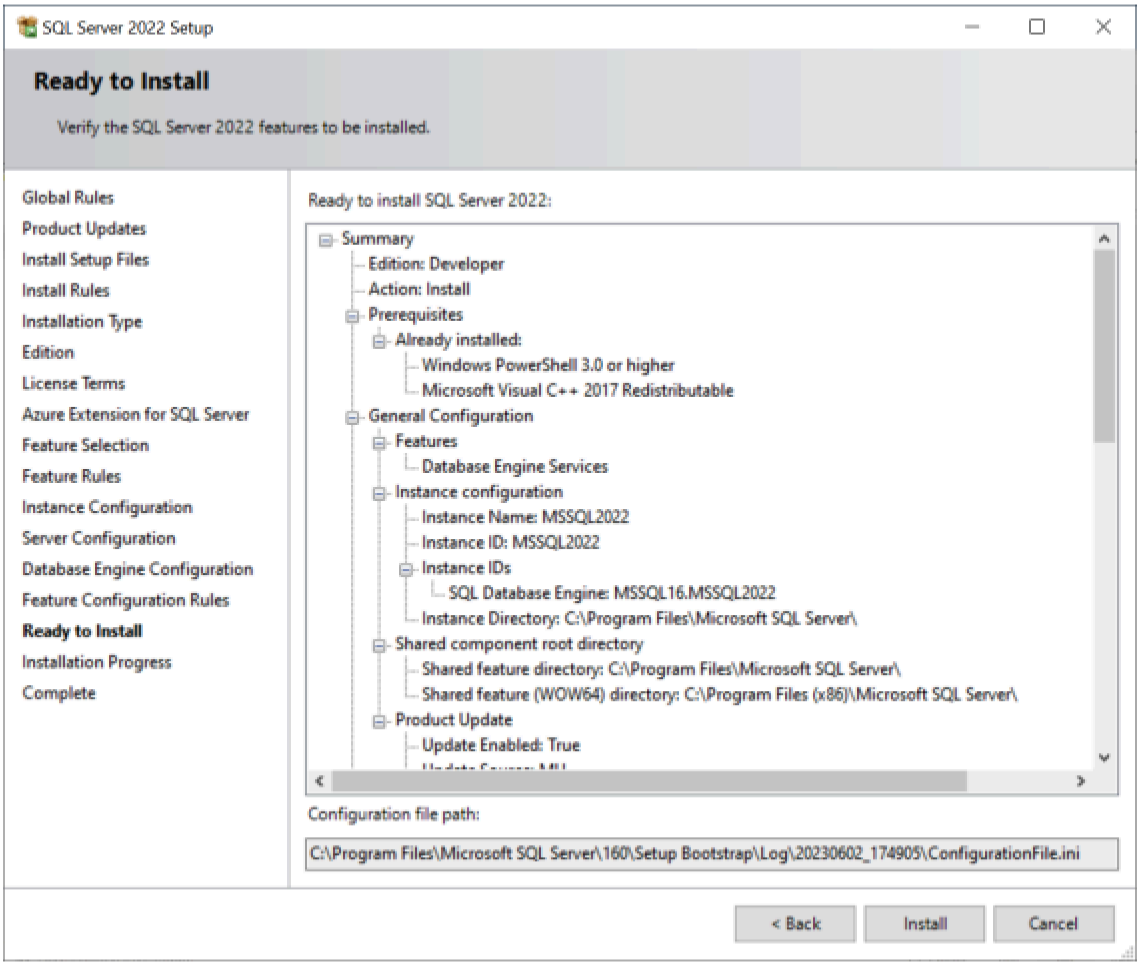
- Click Next button to continue



- Select the Mixed Mode, provide the password for system administration (sa) account (you need to store this password in a secure place so that you can use it to connect to the SQL Server later), click the Add Current User to specify the SQL Server Administrators, and click the Next button



- Verify the SQL Server 2022 features to be installed.



- Click the Close button to complete the installation.

2. Install Microsoft SQL Server Management Studio

To interact with SQL Server, you need to have a SQL Server client tool. Microsoft provides you with the SQL Server Management Studio (SSMS). The SQL Server Management Studio is software for querying, designing, and managing SQL Server on your local computer or in the cloud. It provides you with tools to configure, monitor, and administer SQL Server instances.

- First, download the SSMS from the Microsoft website

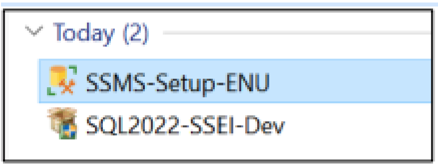

Download SSMS

⤓ Free Download for SQL Server Management Studio (SSMS) 19.1

SSMS 19.1 is the latest general availability (GA) version. If you have a *preview* version of SSMS 19 installed, you should uninstall it before installing SSMS 19.1. If you have SSMS 19.x installed, installing SSMS 19.1 upgrades it to 19.1.

- Release number: 19.1
- Build number: 19.1.56.0
- Release date: May 24, 2023

- Double-click the installation file SSMS-Setup-ENU.exe to launch the SSM installer



- The select the Install button
- Wait for a few minutes while the installer sets up the software
- Once setup is completed, click the Close button
- Now, you should have SQL Server and SQL Server Management Studio installed on
- your computer.
- Use SQL Server Configuration Manager to start or stop SQL Server instance
- OR Start menu > services > MS sqlserver > start/stop service

3.2. SQL Server for Linux or Mac via Docker

Docker is a platform that runs applications in isolated containers, ensuring consistency across environments. Docker Desktop, its user-friendly application, simplifies container management on macOS, Windows, and Linux.

Since SQL Server is not natively supported on macOS, Docker Desktop enables you to run it in a Linux-based container. This setup ensures compatibility, ease of management, and a reproducible environment for development and learning. This guide will show you how to install SQL Server on macOS using Docker.

Step 1: Install Docker Desktop

Follow the link to download and install Docker Desktop: https://docs.docker.com/desktop/setup/install/mac-install/. Make sure that you choose the correct version that is suitable for your chip.
If you are using a Mac with an Apple silicon chip (M1/M2/M3, etc.), it is recommended that you install Rosetta 2. On your Terminal, run the following command:

```
$ softwareupdate --install-rosetta
```

If you are facing the bug that says: " "Docker" is damaged and can't be opened. You should move it to the Bin.".

First, move it to trash, then stop all the Docker processes using Activity Monitor. Then restart your computer before downloading the version 4.37.2 or later using this link.

Step 2: Download and launch the available SQL Server image.

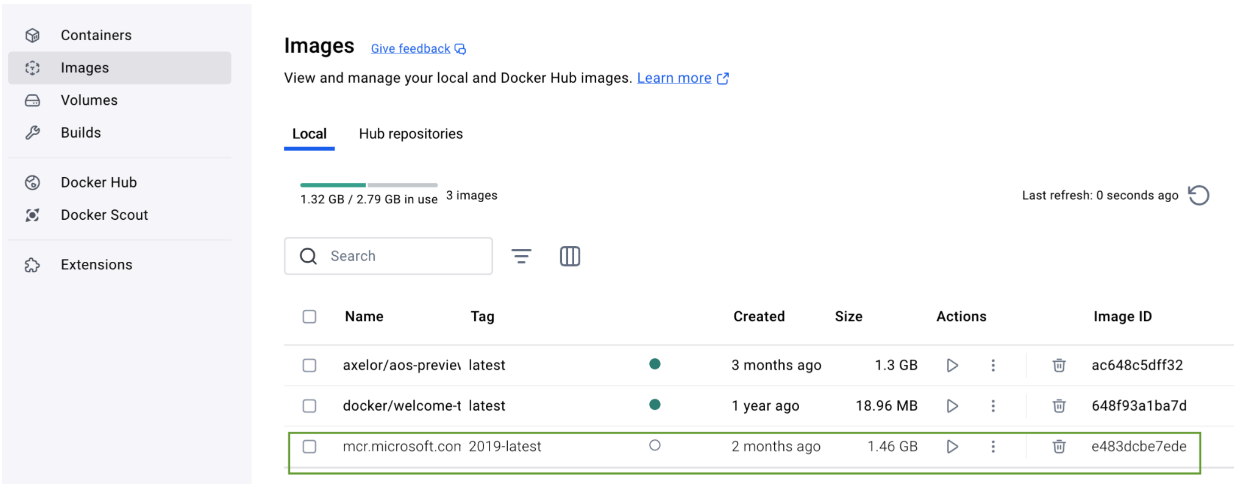
A Docker image is a prepackaged blueprint for running applications in containers. The MSSQL Docker image provides a lightweight, containerized version of Microsoft SQL Server, allowing it to run seamlessly on macOS without native installation.
https://learn.microsoft.com/en-us/sql/linux/quickstart-install-connect-docker?view=sql-server-ver15&pivots=cs1-bash&tabs=cli

Download image

On your Terminal (or Terminal inside the Docker Desktop GUI), run the following commands to pull (download) the Docker image:

```
$ docker pull mcr.microsoft.com/mssql/server:2022-latest
```

If you want to find the image for other versions of MSSQL Server, please check here.
After downloading, you can see the image in the Docker Desktop GUI.



Launch the image

Launch the image to create a docker container. A Docker container is a running instance of a Docker image. Think of the image as a recipe and the container as the prepared dish.

On the Terminal (or Docker Terminal window), run the following command:

```
$ docker run -e "ACCEPT_EULA=Y" -e "MSSQL_SA_PASSWORD=<password>" \
   -p 1433:1433 --name sql_server_test --hostname sql_server_test \
   -d \
   mcr.microsoft.com/mssql/server:2022-latest
```

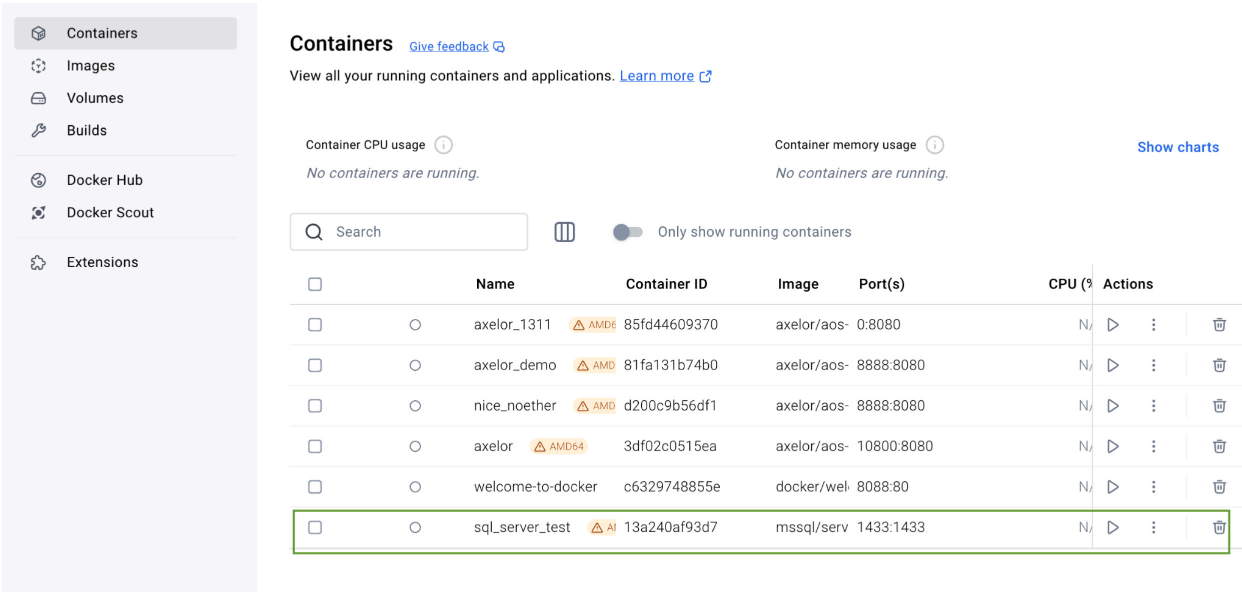
- `-e 'ACCEPT_EULA=Y'`: Confirms by agreeing with end user license agreement (EULA) for Docker.
- `-e 'SA_PASSWORD=<password>'` : Sets the database password.
- `-p 1433:1433` : Map a TCP port on the host environment (first value) with a TCP port in the container (second value). In this example, SQL Server is listening on TCP 1433 in the container and this container port is then exposed to TCP port 1433 on the host.

- `--name sql_server_test` : Sets a name for the Docker container. In this example, we are using "`sql_server_test`"
- `--hostname sql_server_test` : Used to explicitly set the container hostname.
- `-d` : Launches the docker container in daemon mode, allowing it to run in the background without a terminal window open.
- `mcr.microsoft.com/mssql/server:2022-latest` : The SQL server container image

If you're using a Mac with an ARM chip (M1, M2, etc.), you might receive the following warning:

"`WARNING: The requested image's platform (linux/amd64) does not match the detected host platform (linux/arm64/v8) and no specific platform was requested 84182d3f00fb548a5cbf780221417a058b5fe61d3f599d8426eca1d65b5b0dd3`"

However, don't worry your container will still be created after a few seconds. You can check on the Docker GUI (or using docker command: `$ docker ps -l`) for the results

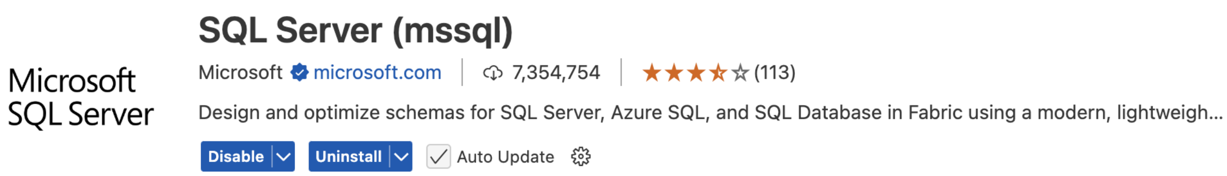


Step 3: Download and install the GUI application

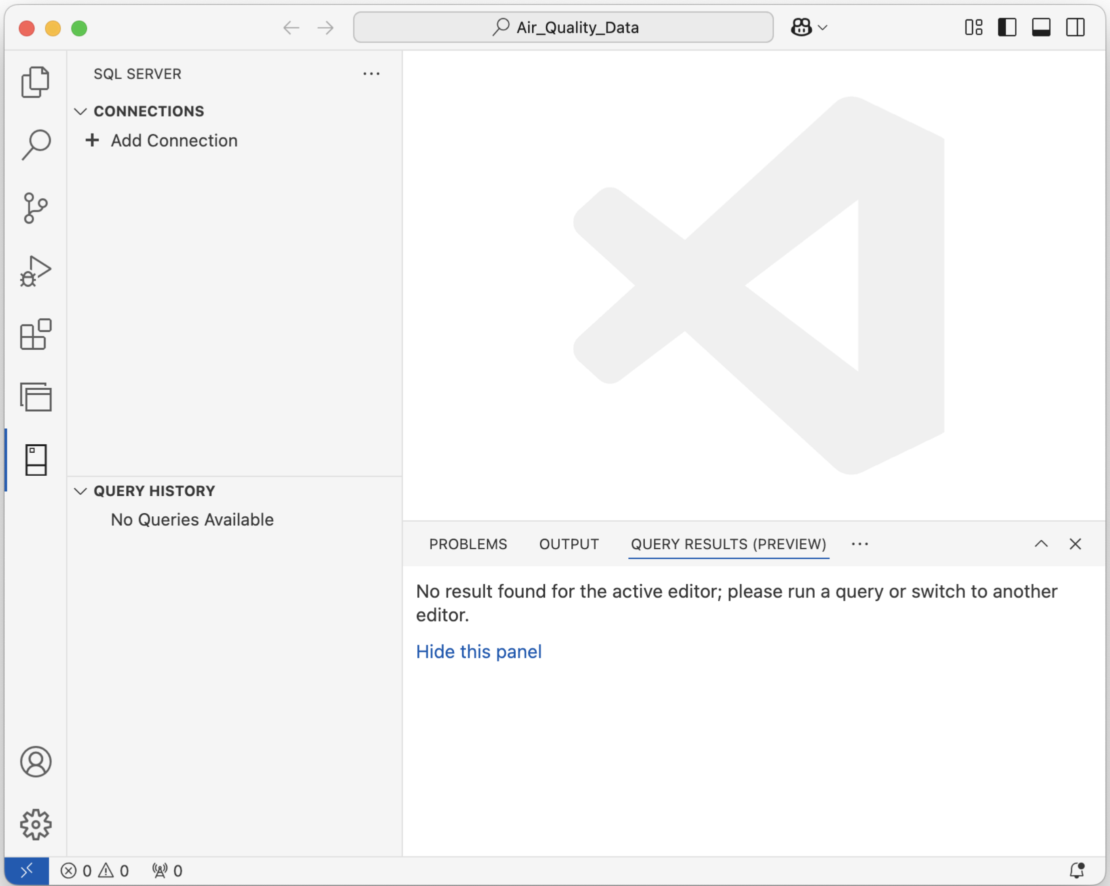
Install SQL Server (mssql) extension on Visual Studio Code

Prerequisites: Visual Studio Code

On your Visual Studio Code UI, navigate to the Extension tabs , search for sql server then install the package.



After installing the extension, you can see the SQL Server Tab at the bottom of VS Code Tab bar , select this tab will navigate to the SQL server GUI



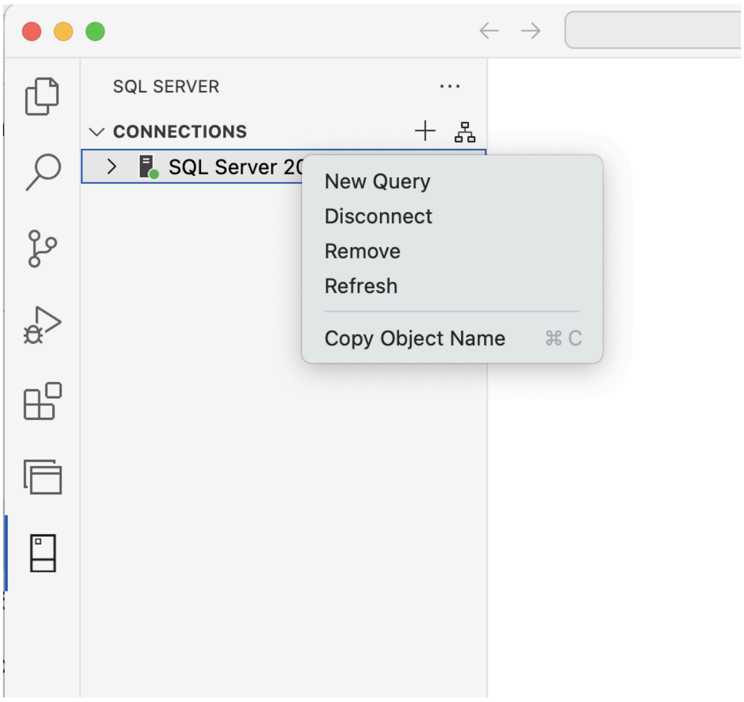
Create new connection to SQL Server

Next, select "+ Add Connection" to create a new connection to your SQL Server. Fill in the form with following information then select "Connect"

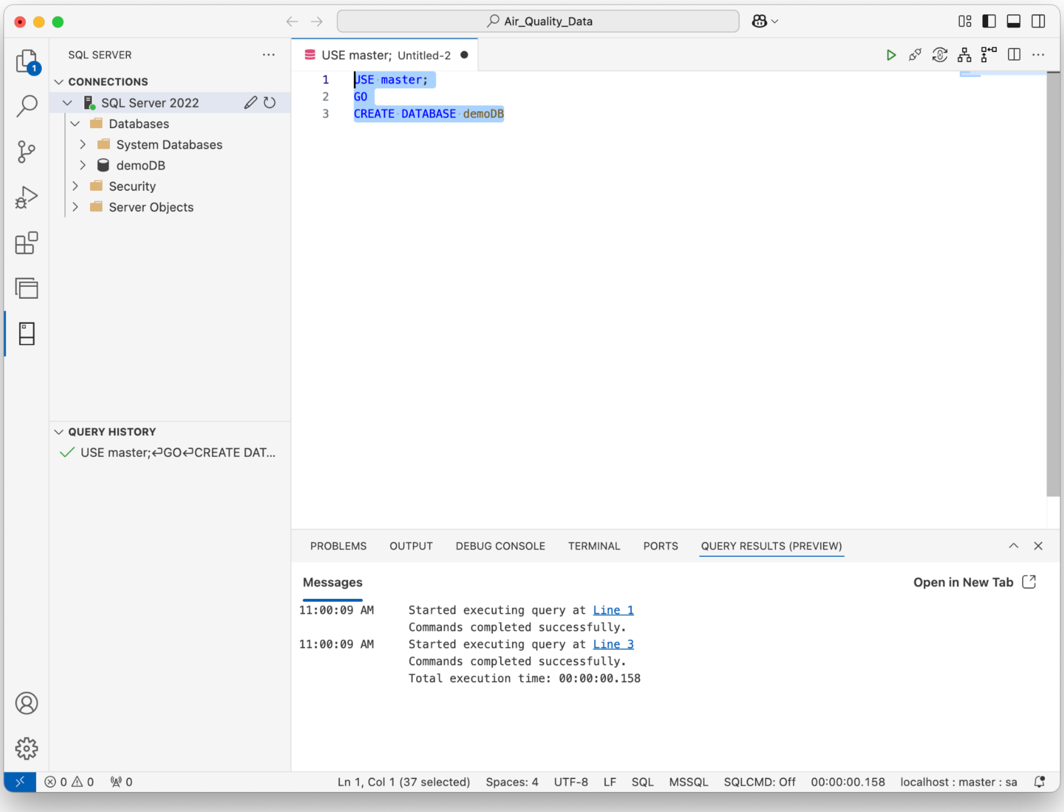
Profile Name: Naming your profile for this connection. For example: SQL Server 2022
Input Type: Paramaters
Server Name: localhost
Trust server certificate: True
Authentication Type: SQL login
User name: sa
Password: <password> the password you specified when launching the image.
Encrypt: Optional

Create your first database

Right click on the Connection, in this example the "SQL Server 20222", choose New query to create a new window to execute SQL queries.



Create new Database:



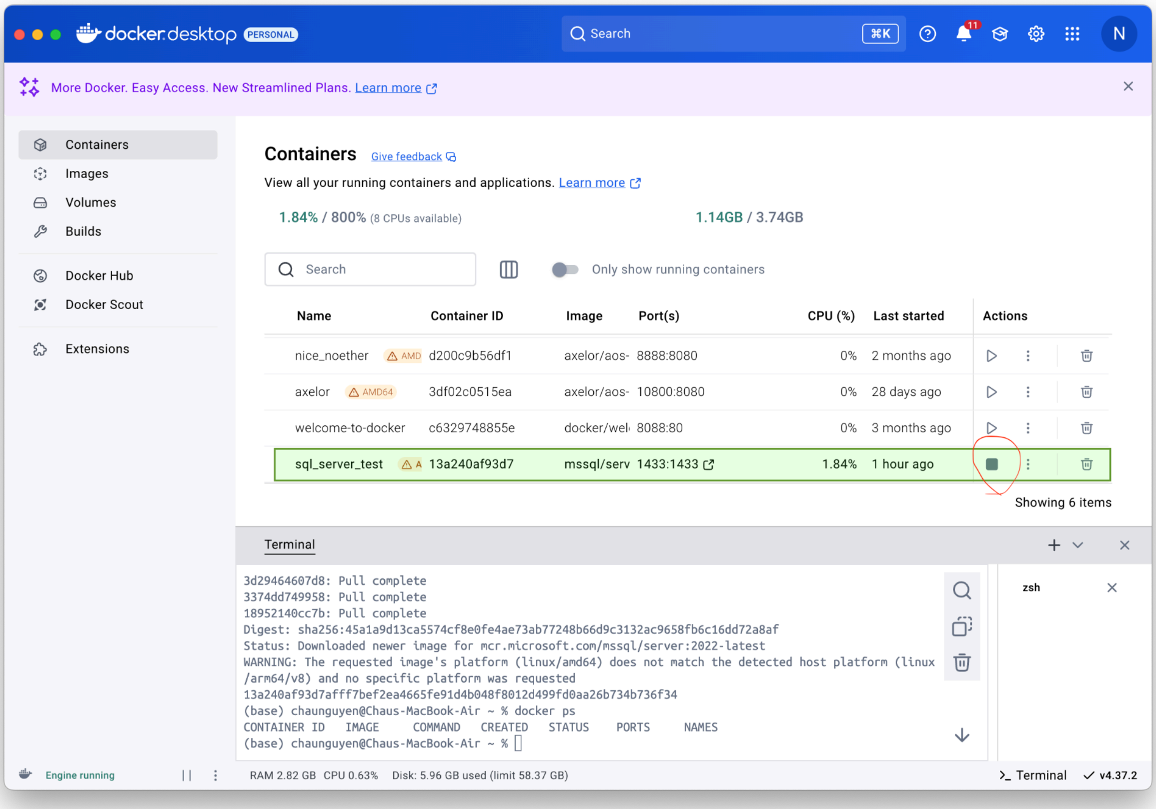
Stop and restart the SQL server

Stop working with SQL Server

When you have finished and saved your work, you can right click on the SQL Connection profile and choose Disconnect to stop the connection with the SQL Server

Then, you should stop your SQL server container as following:

- Using Docker Desktop GUI: Select the stop button of the specific container to stop that. In this example, we stop the SQL server name `sql_server_test`.



- Using Docker commands:
  `$ docker stop sql_server_test`

Restart working with SQL Server container

Restart the SQL server container using Docker GUI and reconnect the connection if you want to work again with SQL server.
Note that you only need to start the container, no need to re-launch the image.

Demo Video

https://www.loom.com/share/c66e6a904c9440b4a3830bfeb951a637?sid=e6a3dec9-fc75-43e2-b232-789baf0c99be

References

1 https://learn.microsoft.com/en-us/sql/linux/quickstart-install-connect-docker?view=sql-server-ver15&pivots=cs1-bash&tabs=cli

2 https://builtin.com/software-engineering-perspectives/sql-server-management-studio-mac